COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31s2
DATE PERFORMED:	06/11/2026	**DATE SUBMITTED:**	06/11/2026
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ACTIVITY NO. 11.1 Containerization

1. PROCEDURE OUTPUT

1. Create a new repository for this activity.

```
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ pwd
/home/jakeworkstation/JakeGestiada/HOA11p1
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ ls
ansible
jakeworkstation@WS:~/JakeGestiada/HOA11p1$
```

Update the packages system

- Sudo apt update && sudo apt upgrade -y

Add docker key GPG key.

- sudo apt install ca-certificates curl
- sudo install -m 0755 -d /etc/apt/keyrings
- sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
- sudo chmod a+r /etc/apt/keyrings/docker.asc

```
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ curl --version
curl 8.5.0 (x86_64-pc-linux-gnu) libcurl/8.5.0 OpenSSL/3.0.13 zlib/1.3 brotli/1.1.0 zstd/1.5.5 libidn2/2.3.7 libpsl/0.21.2 (+libidn2/2.3.7)
libssh/0.10.6/openssl/zlib nghttp2/1.59.0 librtmp/2.3 OpenLDAP/2.6.10
Release-Date: 2023-12-06, security patched: 8.5.0-2ubuntu10.15
Protocols: dict file ftp ftps gopher gophers http https imap imaps ldap ldaps mqtt pop3 pop3s rtmp rtsp scp sftp smb smbs smtp smtps telnet
tftp
Features: alt-svc AsynchDNS brotli GSS-API HSTS HTTP2 HTTPS-proxy IDN IPv6 Kerberos Largefile libz NTLM PSL SPNEGO SSL threadsafe TLS-SRP Un
ixSockets zstd
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ ls -ld /etc/apt/keyrings
drwxr-xr-x 2 root root 4096 Oct  7 15:57 /etc/apt/keyrings
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ ls -l /etc/apt/keyrings/docker.asc
-rw-r--r-- 1 root root 3817 Oct  7 15:57 /etc/apt/keyrings/docker.asc
jakeworkstation@WS:~/JakeGestiada/HOA11p1$
```

Add the repository to Apt sources

- sudo tee /etc/apt/sources.list.d/docker.sources <<EOF
  Types: deb
  URIs: https://download.docker.com/linux/ubuntu
  Suites: (. $/etc/os - release \&\& echo$"{UBUNTU_CODENAME:-$VERSION_CODENAME}")
  Components: stable
  Architectures: $(dpkg --print-architecture)
  Signed-By: /etc/apt/keyrings/docker.asc
  EOF
- Sudo apt update

```
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ sudo tee /etc/apt/sources.list.d/docker.sources <<EOF
>
> Types: deb
> URIs: https://download.docker.com/linux/ubuntu
> Suites: $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}")
> Components: stable
> Architectures: $(dpkg --print-architecture)
> Signed-By: /etc/apt/keyrings/docker.asc
> EOF

Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: noble
Components: stable
Architectures: amd64
Signed-By: /etc/apt/keyrings/docker.asc
```

```
Reading state information... Done
158 packages can be upgraded. Run 'apt list --upgradable' to see them.
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ apt-cache policy docker-ce
docker-ce:
  Installed: (none)
  Candidate: 5:29.8.2-1~ubuntu.24.04~noble
  Version table:
     5:29.8.2-1~ubuntu.24.04~noble 500
        500 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages
     5:29.8.1-1~ubuntu.24.04~noble 500
        500 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages
     5:29.8.0-1~ubuntu.24.04~noble 500
        500 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages
     5:29.7.2-1~ubuntu.24.04~noble 500
        500 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages
     5:29.7.1-1~ubuntu.24.04~noble 500
        500 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages
     5:29.7.0-1~ubuntu.24.04~noble 500
        500 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages
     5:29.6.2-1~ubuntu.24.04~noble 500
        500 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages
     5:29.6.1-1~ubuntu.24.04~noble 500
        500 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages
```

2. Install Docker and enable the docker socket.

Install the Docker Engine

- sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin

```
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ docker --version
Docker version 29.8.2, build 7fc2dff
```

For checking, verify if the docker running is active and start.

- sudo systemctl status docker
- sudo systemctl start docker
- sudo docker run hello-world

```
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ sudo systemctl enable docker.service
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ sudo systemctl enable containerd.service
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ sudo systemctl start docker
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ sudo systemctl status docker --no-pager
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-10-07 16:14:33 PST; 55s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 24590 (dockerd)
      Tasks: 10
     Memory: 45.7M (peak: 45.8M)
        CPU: 932ms
     CGroup: /system.slice/docker.service
             └─24590 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock
```

Enable the docker socket

- sudo systemctl enable docker.socket
- sudo systemctl start docker.socket

Verify

- sudo systemctl status docker.socket --no-pager

```
docker.socket                                 enabled         enabled
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ sudo systemctl enable docker.socket
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ sudo systemctl start docker.socket
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ sudo systemctl status docker.socket --no-pager
● docker.socket - Docker Socket for the API
     Loaded: loaded (/usr/lib/systemd/system/docker.socket; enabled; preset: enabled)
     Active: active (running) since Wed 2026-10-07 16:14:32 PST; 2min 29s ago
   Triggers: ● docker.service
     Listen: /run/docker.sock (Stream)
      Tasks: 0 (limit: 3437)
     Memory: 0B (peak: 256.0K)
        CPU: 768us
     CGroup: /system.slice/docker.socket

Oct 07 16:14:32 WS systemd[1]: Starting docker.socket - Docker Socket for the API...
Oct 07 16:14:32 WS systemd[1]: Listening on docker.socket - Docker Socket for the API.
```

Test docker installation:

- sudo docker run hello-world

```
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ sudo docker run hello-world


Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

3. Add to Docker group to your current user.

- sudo groupadd docker
- sudo usermod -aG docker $USER

```
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ sudo usermod -aG docker $USER
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ getent group docker
docker:x:984:jakeworkstation
```

Apply the Docker group membership

- newgrp docker
- groups

```
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ newgrp docker
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ groups
docker adm cdrom sudo dip plugdev users lpadmin jakeworkstation
```

Test Docker without sudo

- docker run hello-world

```
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

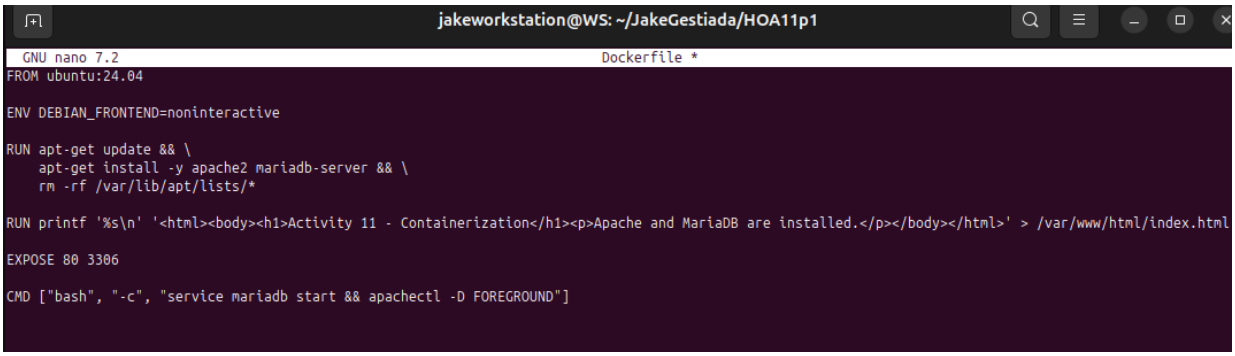
Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

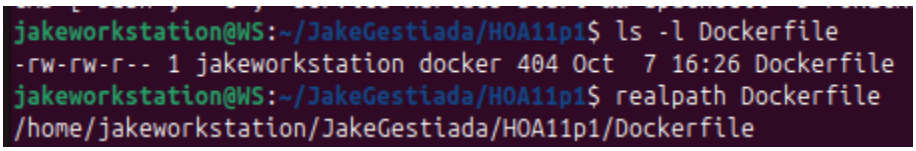
For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

4. Create a Dockerfile to install web and DB server.

Create the Dockerfile

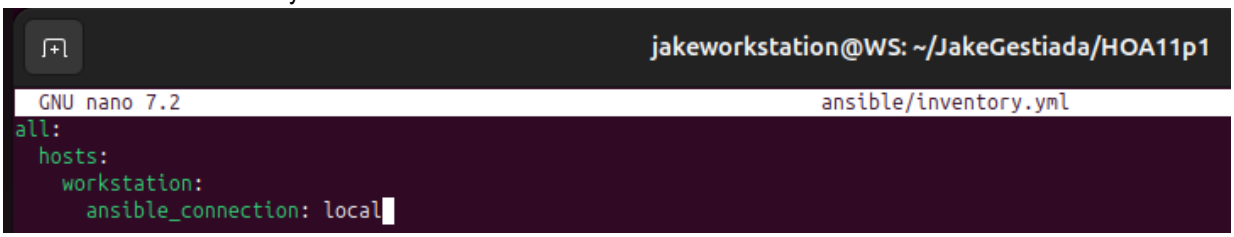
- nano Dockerfile





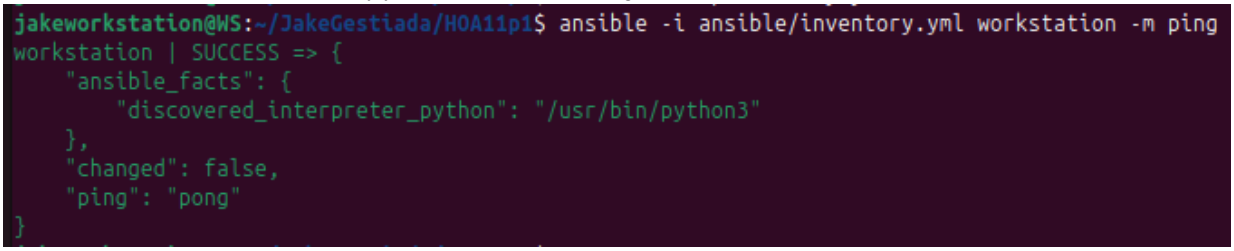
5. Install and build the Dockerfile using Ansible.

Create the Ansible inventory



Test the Ansible connection

- ansible -i ansible/inventory.yml workstation -m ping



Create the Ansible playbook

jakeworkstation@WS: ~/JakeGestiada/HOA11p1

GNU nano 7.2 ansible/site.yml *

```
---
- name: Build Docker image on Workstation
  hosts: workstation
  become: true

  tasks:
    - name: Create Docker build directory
      ansible.builtin.file:
        path: /opt/HOA11p1
        state: directory
        mode: '0755'

    - name: Copy Dockerfile
      ansible.builtin.copy:
        src: ../Dockerfile
        dest: /opt/HOA11p1/Dockerfile
        mode: '0644'

    - name: Build Docker image
      ansible.builtin.command:
        cmd: docker build -t hoa11-web-db:1.0 /opt/HOA11p1

    - name: Remove old container
      ansible.builtin.command:
        cmd: docker rm -f hoa11-web-db
      failed_when: false

    - name: Run Docker container
      ansible.builtin.command:
        cmd: docker run -d --name hoa11-web-db -p 8080:80 -p 3306:3306 hoa11-web-db:1.0
```

Run the Ansible playbook

- ansible-playbook -i ansible/inventory.yml ansible/site.yml

```
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ ansible-playbook -i ansible/inventory.yml ansible/site.yml

PLAY [Build Docker image on Workstation] ***************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
ok: [workstation]

TASK [Create Docker build directory] *******************************************************************************************************************
changed: [workstation]

TASK [Copy Dockerfile] *********************************************************************************************************************************
changed: [workstation]

TASK [Build Docker image] ******************************************************************************************************************************
changed: [workstation]

TASK [Remove old container] ****************************************************************************************************************************
changed: [workstation]

TASK [Run Docker container] ****************************************************************************************************************************
changed: [workstation]

PLAY RECAP *********************************************************************************************************************************************
workstation                : ok=6    changed=5    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Verifying Docker Image, Running Container, Apache Web Server, MariaDB, and Port Mappings

```
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ docker image ls
                                                                                                    i Info →  U In Use
IMAGE               ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest  5e2309035332       25.9kB         9.49kB    U
hoa11-web-db:1.0    c8f586e85e38        660MB          122MB    U
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ docker ps
CONTAINER ID   IMAGE             COMMAND                  CREATED         STATUS         PORTS
                    NAMES
17a5011f3cdf   hoa11-web-db:1.0   "bash -c 'service ma…"   9 minutes ago   Up 9 minutes   0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 0.0.0.0:8080->80/tc
p, [::]:8080->80/tcp   hoa11-web-db
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ curl http://localhost:8080
<html><body><h1>Activity 11 - Containerization</h1><p>Apache and MariaDB are installed.</p></body></html>
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ curl -I http://localhost:8080
HTTP/1.1 200 OK
Date: Wed, 07 Oct 2026 08:44:01 GMT
Server: Apache/2.4.58 (Ubuntu)
Last-Modified: Wed, 07 Oct 2026 08:33:41 GMT
ETag: "6a-65d3bf8ed0340"
Accept-Ranges: bytes
Content-Length: 106
Vary: Accept-Encoding
Content-Type: text/html

jakeworkstation@WS:~/JakeGestiada/HOA11p1$ docker exec hoa11-web-db service mariadb status
 * /usr/bin/mariadb-admin  Ver 10.0 Distrib 10.11.14-MariaDB, for debian-linux-gnu on x86_64
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Server version          10.11.14-MariaDB-0ubuntu0.24.04.1
Protocol version        10
Connection              Localhost via UNIX socket
UNIX socket             /run/mysqld/mysqld.sock
Uptime:                 10 min 7 sec

Threads: 1  Questions: 61  Slow queries: 0  Opens: 33  Open tables: 26  Queries per second avg: 0.100
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ docker port hoa11-web-db
80/tcp -> 0.0.0.0:8080
80/tcp -> [::]:8080
3306/tcp -> 0.0.0.0:3306
3306/tcp -> [::]:3306
```

Structure form:

```
jakeworkstation@WS:~/JakeGestiada/HOA11p1$ tree
.
├── ansible
│   ├── inventory.yml
│   └── site.yml
└── Dockerfile
```

Open the web server in the browser



6. Add, commit and push it to your repository
https://github.com/DamienJakeGestiada/CPE212_JakeGestiada/tree/main/HOA11p1

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:
Answer the following:
1. What are the benefits of implementing containerizations?
   - Containerization helps in the ease of installation, running, and management of applications as they come together in one container with all their needed components. Besides, it isolates applications; it runs with the use of less resources compared to traditional virtual machines. Containerization also ensures the speed of deployment and consistency of deployment. Docker let us run web and database servers in the same environment during the activity whereas Ansible was used to automate the setup and deployment.

Conclusion:
- The HOA shows the potential of Docker in launching and maintaining containers for various functionalities such as a web server and a database server. We wrote a Dockerfile script that would install Apache and then a separate one that would be used to install MariaDB. After that, with Ansible, a workflow was developed that would handle automatically all the steps related to the assembling and running the Docker container. Also, the status of the container, Apache, and MariaDB were checked. Besides that, it gave us hands-on experience in knowing that the concept of containerization can help one in deployment as well as it is a more organized form of deployment. Generally, the practice emphasized the combination of Docker and Ansible to make server management and application deployment more straightforward and effortless.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini and Grammarly**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **For grammar correction and missing code input.**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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